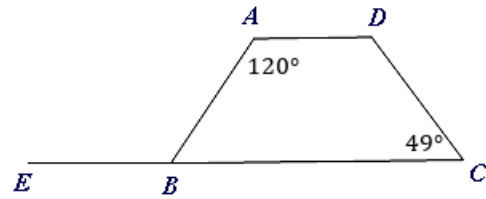


Properties of trapezoids



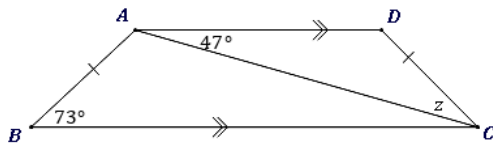
1 Consider the diagram shown, given that $ABCD$ is a trapezoid.

- a Identify which sides are parallel.
- b Find $m\angle ABC$.
- c Find $m\angle ADC$.

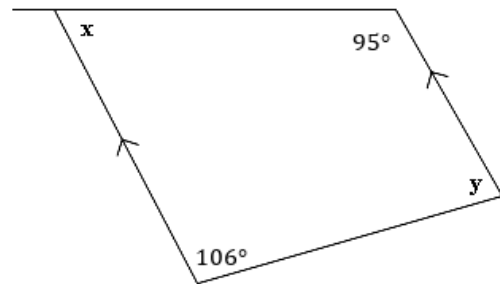


2 Find the value of the variable(s) in the following diagrams.

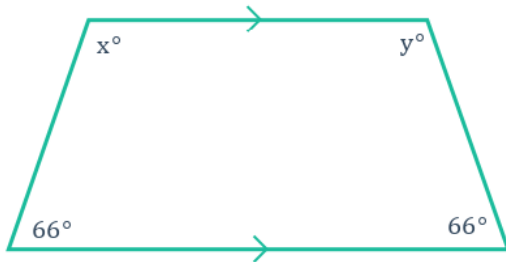
a



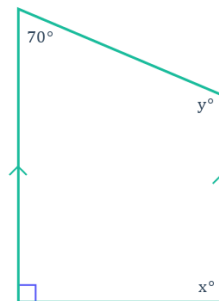
b



c



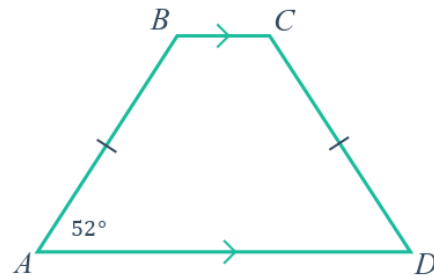
d



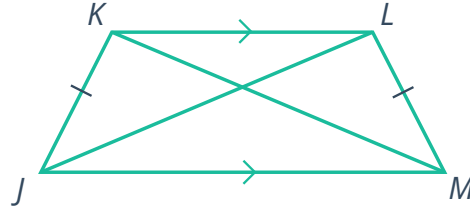
3 Consider the following trapezoid.



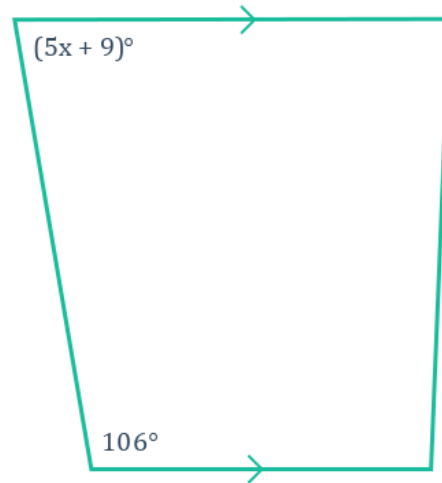
- a Find the value of $m\angle B$.
- b Find the value of $m\angle C$.
- c Find the value of $m\angle D$.



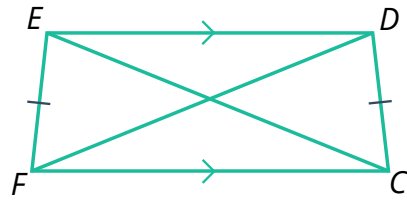
4 Consider the following trapezoid $JKLM$,
where $KM = 20$.
Find the value of JL .



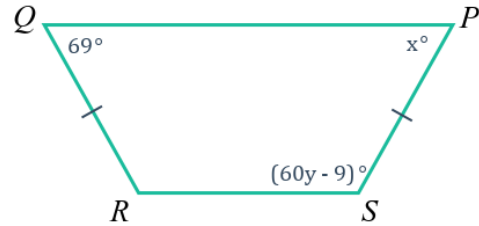
5 Consider the following trapezoid.
Find the value of x .



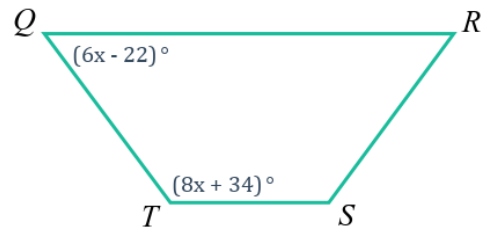
- 6 Consider the following trapezoid $CDEF$.
For this trapezoid, $EC = 21$ and
 $FD = 6x - 9$.
Find the value of x .



- 7 Consider the following trapezoid.
- Find the value of x .
 - Find the value of y .



- 8 Consider the following trapezoid.
- Find the value of x .
 - Find $m\angle Q$.



- 9 Given: $\overline{AB} \parallel \overline{DC}$, $\overline{AD} \cong \overline{BC}$, $\overline{AD} \parallel \overline{BE}$
 Prove: $\angle BCE \cong \angle ADE$ and
 $\angle BAD \cong \angle ABC$
 Note that D , E , and C are collinear.



To prove: $\angle BCE \cong \angle ADE$ and $\angle BAD \cong \angle ABC$

Statements	Reasons
1. $\overline{AB} \parallel \overline{DC}$, $\overline{AD} \cong \overline{BC}$, $\overline{AD} \parallel \overline{BE}$	Given
2. $ABED$ is a parallelogram	Definition of a parallelogram
3. $\overline{AD} \cong \overline{BE}$	Opposites sides of a quadrilateral are congruent
4. $\overline{BE} \cong \overline{BC}$	Transitive property of congruence
5. $\triangle EBC$ is isosceles	Definition of an isosceles triangle
6. $\angle BCE \cong \angle BEC$	Base angles of an isosceles triangle are congruent
7. $\angle BEC \cong \angle ADE$	Corresponding angles theorem
8. $\angle BCE \cong \angle ADE$	Transitive property of congruence
9. $\angle BCE$ and $\angle ABC$ are supplementary	Consecutive interior angles postulate
10. $\angle ADE$ and $\angle BAD$ are supplementary	Consecutive interior angles postulate
11. $\angle BAD \cong \angle ABC$	Congruent supplements theorem

Given the proof above, determine whether the following statements are true.

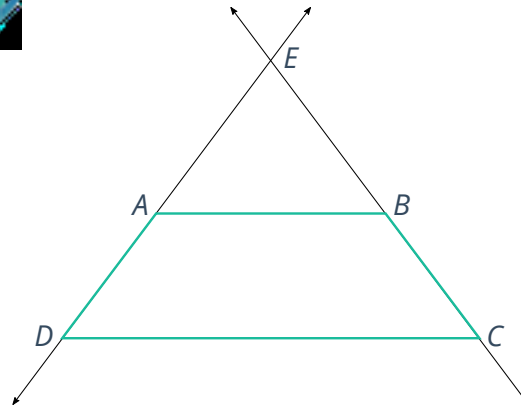
- a If a trapezoid is isosceles, then base angles are congruent.
- b If a trapezoid has one pair of congruent base angles, then it is isosceles.
- c If a quadrilateral is a trapezoid, then base angles are supplementary.
- d If a quadrilateral has one pair of supplementary base angles, then it is a trapezoid.

10 Given:



- $ABCD$ is a trapezoid with bases $\overline{AB}$ and $\overline{DC}$
- $\angle BCD \cong \angle ADC$
- $\overleftrightarrow{AD}$ and $\overleftrightarrow{BC}$ intersect at E

Prove: $ABCD$ is an isosceles trapezoid



To prove: $ABCD$ is an isosceles trapezoid

Statements	Reasons
1. $ABCD$ is a trapezoid	Given
2. $\angle BCD \cong \angle ADC$	Given
3. $\overline{AB} \cong \overline{DC}$	Definition of a trapezoid
4. $\angle EAB \cong \angle EDC$ and $\angle EBA \cong \angle ECD$	Corresponding angles theorem
5. $\angle EAB \cong \angle EBA$	Transitive property of congruence
6. $\triangle EAB$ is isosceles and $\triangle EDC$ is isosceles	If a triangle has congruent base angles, then it is isosceles.
7. $\overline{AE} \cong \overline{BE}$ and $\overline{DE} \cong \overline{CE}$	Definition of an isosceles triangle
8. $AE = BE$ and $DE = CE$	Definition of congruence
9. $DA + AE = DE$ and $CB + BE = CE$	Segment addition postulate
10. $DA = DE - AE$ and $CB = CE - BE$	Subtraction property of equality
11. $DA = CB$	Substitution
12. $\overline{DA} \cong \overline{CB}$	Definition of congruence

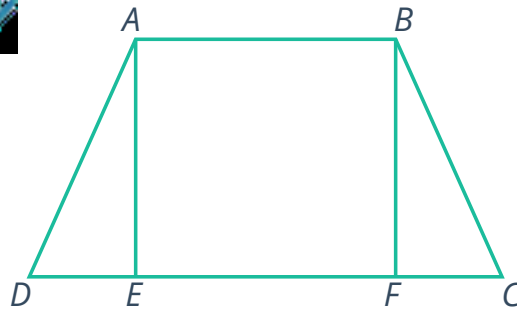
Given the proof above, determine whether the following statements are true.

- a If a trapezoid has congruent legs, then it is isosceles.
- b If a trapezoid is isosceles, then it has congruent legs.
- c If a trapezoid has one pair of congruent base angles, then it is isosceles.
- d If a trapezoid is isosceles, then it has one pair of congruent base angles.

- 11 Given that $ABCD$ is an isosceles trapezoid with altitudes $\overline{AE}$ and $\overline{BF}$.



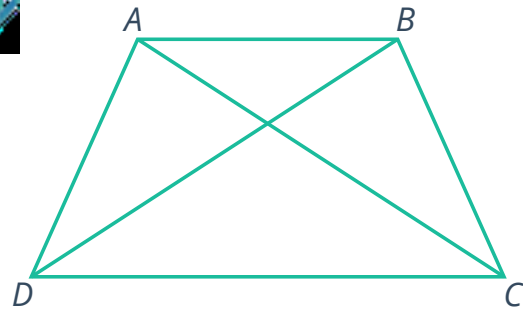
Complete the following two-column proof that proves that each pair of base angles are congruent.



To prove: Each pair of base angles of trapezoid $ABCD$ are congruent.

Statements	Reasons
1. $ABCD$ is an isosceles trapezoid	Given
2. $\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3. $\overline{AE} \cong \overline{BF}$	Altitudes of a trapezoid are congruent
4. $\angle DEA$ and $\angle CFB$ are right angles	Definition of an altitude
5. $\angle DEA \cong \angle CFB$	All right angles are congruent
6. $\parallel$	$\parallel$
7. $\parallel$	$\parallel$
8. $\angle BCD$ and $\angle CBA$ are supplementary	Consecutive interior angles postulate
9. $\angle ADC$ and $\angle BAD$ are supplementary	Consecutive interior angles postulate
10. $\angle BAD \cong \angle ABC$	Congruent supplements theorem

- 12 Given that $ABCD$ is an isosceles trapezoid with bases $\overline{AB}$ and $\overline{DC}$. Complete the following two-column proof that proves that $\overline{AC} \cong \overline{BD}$.



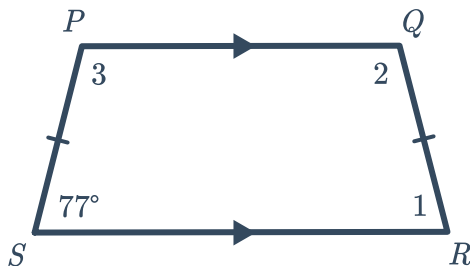
To prove: $\overline{AC} \cong \overline{BD}$

Statements	Reasons
1. $ABCD$ is an isosceles trapezoid	Given
2. $\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3. $\parallel$	$\parallel$
4. $\angle ADC \cong \angle BCD$	If a trapezoid is isosceles, then each pair of base angles is congruent.
5. $\triangle ADC \cong \triangle BCD$	Side-angle-side congruence theorem
6. $\overline{AC} \cong \overline{BD}$	Corresponding parts of congruent triangles are congruent (CPCTC)

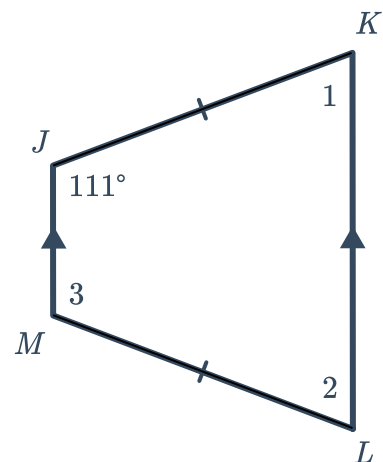
Additional questions

- 13 Solve for each of the missing angle measures in the following trapezoids:

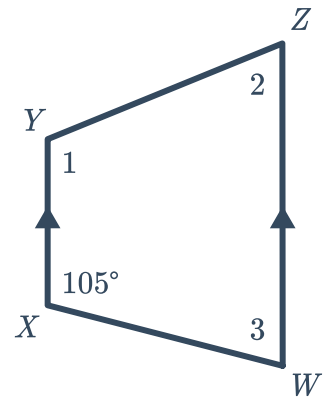
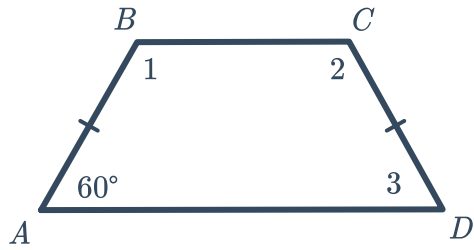
a



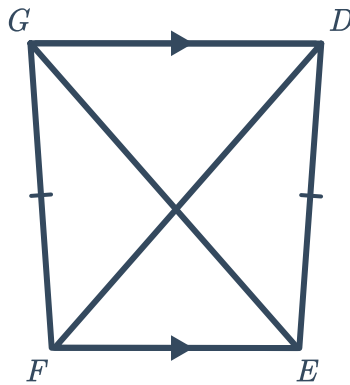
b



c

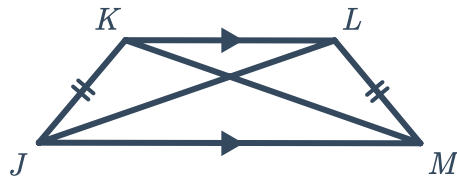


- 14 Given: Trapezoid $DEFG$
 $EG = 3.5$



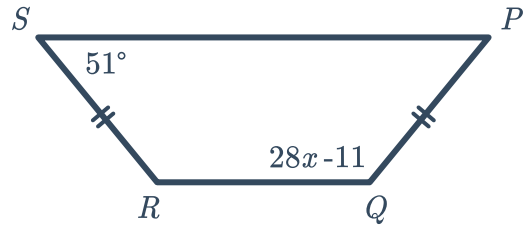
Solve for FD .

- 15 Given: Trapezoid $JKLM$
 $KM = 1.26$



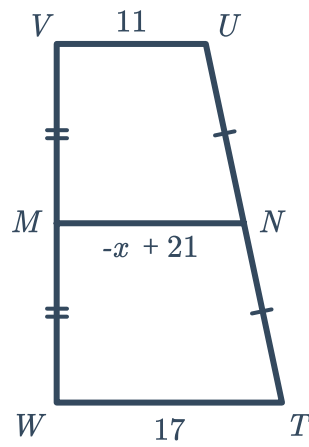
Solve for JL .

16 Given: Trapezoid $PQRS$



Solve for x .

17 Given: Trapezoid $TUVW$

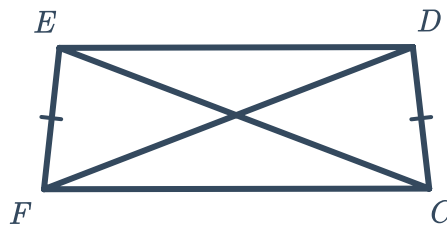


Solve for x .

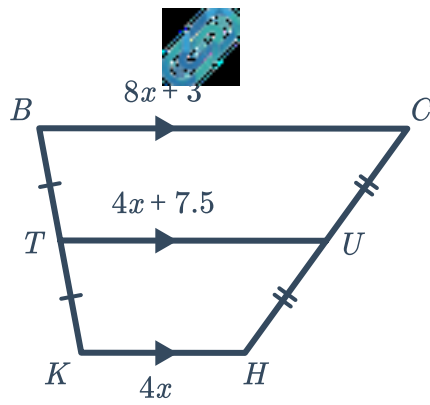
18 Given: Trapezoid $FEDC$

$$EC = 4x + 11$$

$$FD = 3x + 17$$

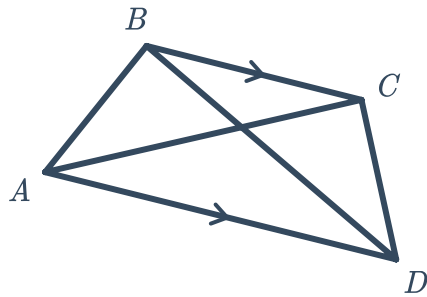


Solve for x .



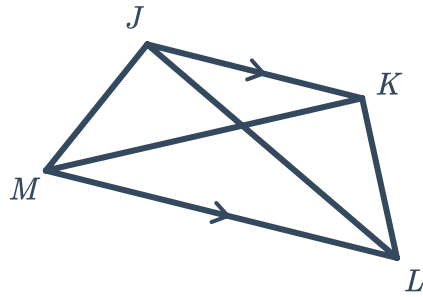
- a Solve for x
- b Solve for BC
- c Solve for TU
- d Solve for KH

20 Given: $\angle BAD = (5x^2 + 32)^\circ$
 $\angle CDA = (7x^2)^\circ$

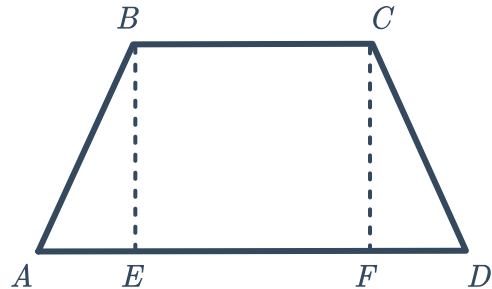


Find the value of x that makes Trapezoid $ABCD$ isosceles.

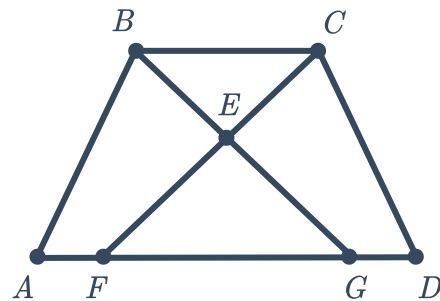
- 21 Determine whether each of the following statements about isosceles trapezoid $JKLM$ is true or false and explain how you know.



- a $\overline{JM} \cong \overline{KL}$
 - b $\overline{JL} \cong \overline{KM}$
 - c $\overline{JK} \cong \overline{ML}$
 - d $\overline{JM} \parallel \overline{KL}$
 - e $\angle J \cong \angle L$
 - f $\angle K$ and $\angle M$ are supplementary.
- 22 Given: Trapezoid $ABCD$ with altitudes $\overline{BE}$ and $\overline{CF}$
 $\overline{AB} \cong \overline{CD}$
 To prove: $\angle A \cong \angle D$



- 23 Given: Trapezoid $ABCD$
 $\overline{EF} \cong \overline{EG}$
 $\overline{AF} \cong \overline{DG}$
 $\overline{BG} \cong \overline{CF}$
 To prove: Trapezoid $ABCD$ is isosceles



- 24** Given: Trapezoid $ABCD$ is isosceles
 $\angle BAK \cong \angle BKA$
To prove: $BKDC$ is a parallelogram

